

Supplementary Online Content

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RESOLVE Study Description of Interventions based on Template for Intervention Description and Replication (TIDieR)

This supplementary material has been provided by the authors to give readers additional information about their work

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	Usual Care Plus	painTRAINER	Health Coach
WHY Goal and rationale of the elements essential to the intervention	Goal: To allow participants to pursue pain management-related care as they normally would within their healthcare system. Rationale: Usual care comparison enables assessment of benefit of the study interventions over what treatments participants are already able to access to help manage their pain. This is important to understand from a clinical and healthcare-implementation (cost) perspective. The resource manual (“plus”) provided to participants in this group was consistent with educational materials for healthcare self-management available to patients in participating healthcare systems.	Goal: To provide an online standardized cognitive behavioral therapy for chronic pain (CBT-CP)-based skills training program designed to accommodate low-literacy participant needs and allow individuals a flexible approach to skills training they can tailor (time/skill emphasis) to their needs. The online format allowed a largely self-directed approach provided at minimal expense (i.e., not dependent on clinician involvement and support limited to technical help and engagement outreach). Rationale: Self-directed CBT-CP-based skills training allows for a flexible, low-cost approach that lowers barriers for high quality CBT-CP-based skills training for those with online access. Multi-media features provided in an easy-to-use format utilizing computerized algorithms (based on expert clinician guidance) to tailor participant experience and designed to enhance development and sustained use of pain management coping skills.	Goal: To provide standardized cognitive behavioral therapy for chronic pain (CBT-CP)-based skills training with a focus on tailoring the specific elements of the skills training approach to participants’ circumstances in real time. Virtual delivery lowers barriers for receipt of services for those in rural/underserved regions of the country or otherwise unable to access clinical providers skilled in evidence-based behavioral pain services. Interaction with a trained therapist to enhance connectedness and accountability. Rationale: CBT-CP-based program delivered by centrally located, master's-level clinicians should enable reinforcement of progress, tailored help for participants in identifying and problem-solving obstacles, while retaining benefits of standardized training, treatment adherence, and scalability. Further, nonspecific elements of therapist contact may enhance therapeutic outcomes.
WHAT			
Participant Materials	<i>American Chronic Pain Association Resource Guide to Chronic Pain Management</i>⁷, 2020 edition This educational guide provides comprehensive information describing the spectrum of pain management modalities, including passive, active and self-directed interventions such as medications, nutrition, exercise, complementary, alternative and integrative medicine, and biopsychosocial strategies. It is intended to educate individuals and does not advise or instruct on treatments but can	<u>Participant Workbook:</u> Mailed to participant prior to first contact from study staff. Includes instructions for registering, unique program login information, and details on “How to Use the Program,” with screenshots and text describing the website’s structure, features, how to use and outlines of session content. <u>Program Website:</u> (mypaintrainer.org) Online sessions that must be completed in a pre-determined order, each teaching one or more evidence-based pain coping skill⁸; led by a recorded “virtual coach” (audio to minimize reading and support needs for low-literacy	<u>Participant Workbook:</u> Mailed to participant prior to first contact from health coach. Includes overview of each session (content on the concepts and skills taught), worksheets to help aid learning of concepts and to log skills practices each week. <u>Participant Resources Website:</u> - 4 audio recordings of guided exercises lasting 5-20 min. (Breath Awareness, 3-Part Breath, Progressive Muscle Relaxation, Guided Imagery)

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	help inform patients on treatment decisions to be made in conjunction with and approved by their healthcare provider.	participants). Provides training in an easy-to-use format that includes animated demonstrations, interactive exercises, tips for working through common barriers to behavior change/skills practice, and tailored feedback to reinforce learning.	- 2 instructional videos lasting 5 min. each (Science of Thinking, Feeling and Responding to Pain; Activity / Rest Cycle) - Access to electronic (PDF) version of Workbook materials by session
Staff & Interventionist Materials	None	Scripts for staff regarding onboarding and engagement calls	Health coaches used Standardized Treatment Manual (“Health Coach Manual”)
Procedures	Printed / hard copy of the resource guide mailed to participants	Eight participant self-completed online sessions, 30-45 minutes each Participant workbook guided planning and goal setting	Participants had Eight one-on-one sessions with health coach, 45-60 minutes each Participant workbook with information on skills practice and goal setting
WHO PROVIDED			
Expertise/ Background	Not applicable	Technical support provided by research staff with prior administrative experience working directly with study participants intended to emulate what could be delivered by Medical Assistant type outreach/support.	Health coaches were master's-level behavioral health practitioners with no prior training in chronic pain management (n=12). Centralized team based at 2 clinical study sites. Ongoing supervision provided by clinical psychologist via weekly hour-long team meetings.
Training	Not applicable	Research staff who provided technical support were given approximately 9 hours of training: - 1-hour videoconference with painTRAINER program developer, Christine Rini; provided overview of program registration and structure - 3-hour online training on Motivational Interviewing (MI) techniques provided to KP employees - 1-hour videoconference with study PI on utilizing MI techniques during onboarding and engagement calls - 2 hours of onboarding and engagement call script role play - 2-hour videoconference with study project manager on specific procedures for	Health coaches had approx. 34 hours didactic and experiential coursework: - 90-minute overview of pain psychology - For each session the coursework involved: 1) detailed review and discussion of content in treatment manual and participant workbook; 2) 1-hour, web-based, synchronous didactic training; 3) role playing a participant with clinical supervisor or other health coach; 4) role play as health coach - Clinical supervisors then reviewed for proficiency at least 16 sessions completed with real participants (2 instances for each of the 8 sessions); proficiency required 100% fidelity in delivering session content and 80% fidelity on core CBT skills (i.e., ratings of

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		conducting onboarding and engagement calls, i.e., logging contacts in study tracking system Research staff also given the opportunity to complete each of the painTRAINER online modules.	“adequate/good” or higher for each of the 9 domains on the modified <i>Cognitive Behavioral Therapy Skills Training for Chronic Pain Therapist Rating Scale</i> relevant for a particular session)
HOW	None	- Participants access painTRAINER via website - Participant has unique account with username and password	- Participant completed sessions by phone or videoconference - Sessions were one-on-one with health coach
WHERE	Not applicable	Participant chose location of preference (home, work, etc.) from which to complete online sessions Internet access required; format optimized by computer (vs. smart phone) interface	Participant chose private location of preference (home, work, etc.) from which to complete sessions Phone access required; internet access for videoconferencing optional according to participant preference
WHEN and HOW MUCH	Not applicable	Participants received: - One onboarding and registration call (approx. 15 min) - Access 8 weekly online modules (30-45 min) to be completed during 12-week treatment period - Tools to help do skills practice on own between sessions (daily practice encouraged) - (If applicable) Phone outreach/contact by study staff (MI-based “engagement calls”) if not adhering to the recommended completion schedule: > 7 days since participant registered and session 1 not initiated; initiated a session but incomplete after > 7 days; >10 days since any session completion date; and sessions completed too quickly (3 or more sessions completed in 9 days). N=356 of the 776 in this study group completed ≥1 outreach call with a staff	Participants received: - One onboarding and initial scheduling call (approx. 10 min) - 8 weekly sessions with health coach (45-60 min) to be completed during 12-week treatment period; In limited instances to allow for extenuating circumstances a four-week extension was allowed (n=14) - Encouragement to do skills practice on own between sessions (daily practice encouraged)

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		member (45.9%). Average number of outreach calls completed for these N=356 was 1.8 (Range: 1-7) with an average length of 4 minutes. - Full 12 month access to PainTRAINER online program	
TAILORING/ PERSONAL- IZATION	Participants chose to pursue usual care according to their needs and preferences.	Participants chose when and which skills/strategies to practice. Participants chose to pursue usual care according to their needs and preferences.	Participants chose to complete sessions by phone or videoconferencing. Participants chose when and which skills/strategies to practice. Participants chose to pursue usual care according to their needs and preferences. Health coaches tailored individual participant home practice goals to participants resources, interests, and values. Pain education was tailored to participants' specific musculoskeletal pain condition(s) and related comorbidities.
MODIFICATIONS	None	Added motivational interviewing-based language to the onboarding and registration call script used by research staff to improve engagement. Change was implemented on 8/9/2021 after ~140 participants were enrolled in the study.	None
HOW WELL / FIDELITY			
Planned	Not applicable	Not applicable; treatment is standardized because the intervention is delivered via an online program thus no fidelity monitoring of intervention administration is warranted.	Sessions were audio or video-recorded with participant permission. Health coaches completed self-assessment of their fidelity to session content after each session. - For first 12 months after reaching proficiency, random sample of 5% of each health coach's completed sessions were reviewed by clinical supervisors. Feedback provided electronically with option for 1:1 discussion.

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			- For months 13+, one randomly selected session per coach every 6 months was reviewed for fidelity. Feedback provided electronically with option for 1:1 discussion. If reviewed session did not meet criteria of 100% fidelity in session content and 80% fidelity on CBT treatment principles, additional sessions reviewed, and feedback provided until met proficiency.
Actual	Not applicable	Not applicable	A total of N=299 sessions out of 4,626 completed were reviewed for fidelity (6.5%) - Training to proficiency period: 193 sessions reviewed - First 12 months: 89 sessions out of 1,791 reviewed (5%) - Month 13+ (1 session reviewed every 6 months per coach): 17 sessions out of 2,350 There were N=8 instances in which session content did not meet 100% fidelity and needed to be redone (8/299=2.7%) and N=11 instances where the fidelity to CBT treatment principles did not meet 80% (11/299=3.7%).

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